

Our solar system

In this chapter, you will learn about:

- Stars and planets
- The solar system
- Natural satellites in the solar system

All the students will be able to:

- ✓ Describe the solar system and its planetary arrangement showing position of the Earth in our solar system.
- ✓ Explain that Sun is a star.
- ✓ Differentiate between a star and a planet.
- ✓ Categorize planets and explain the relative size of the planets and their distance from the Sun using a model.
- ✓ Compare the size of the Sun, Earth and Moon.
- ✓ Investigate the moons of the different planets of the solar system.
- ✓ Compare Meteorites, Asteroids and Comets

Have you ever thought about other planets like our planet Earth? Our Sun and objects (planets, comets, asteroids) orbiting the Sun comprise of a family. Do you know the place of Earth, in this family? Have you ever seen a meteor shower or a comet and thought about what are they made of and where did they come from? What is the origin of our Sun and the planets, and why are they so much different in size and appearance?

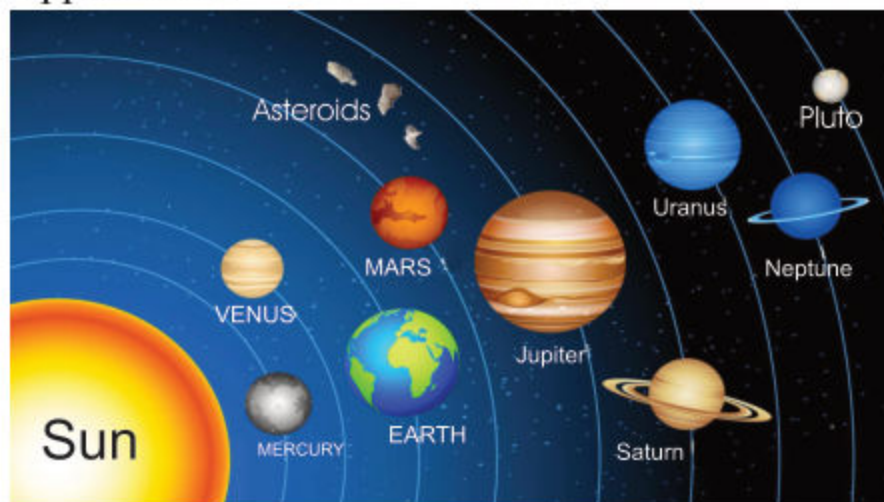


Figure 10.1: Size and distance of planets from the Sun

Do you know what is the name of the national space agency of Pakistan and its functions?

Pakistan Space and Upper Atmosphere Research Commission (SUPARCO) is the national space agency of Pakistan, established in 1961. The main objectives of SUPARCO are,

- to undertake research in the field of science and engineering.
- to utilize the benefits of space technology for the advancement of our country.

Besides these, SUPARCO's main functions are:

- to make satellites for communication, the Earth's observation and research purposes
- to conduct studies in space and atmospheric sciences, radio wave propagation, astronomy and geomagnetism.

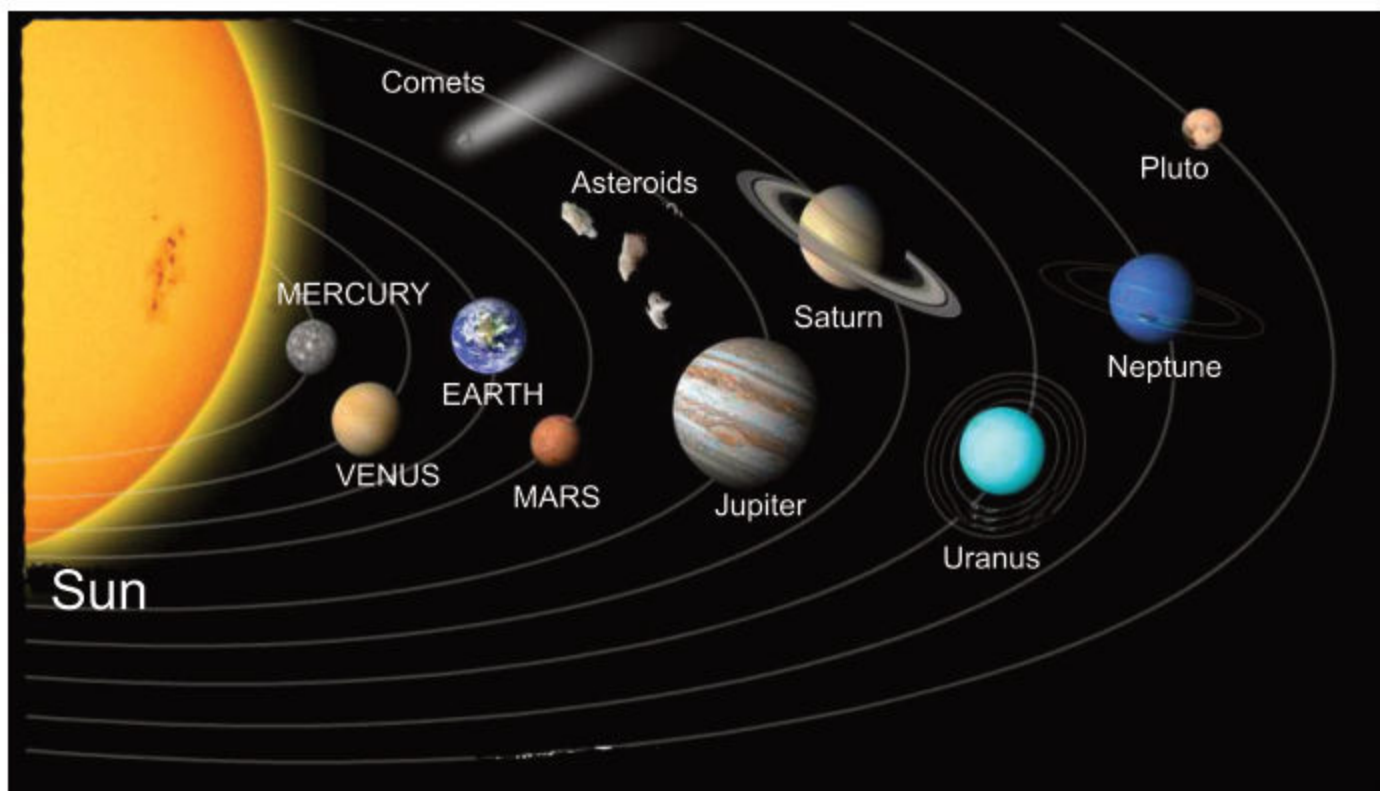
The solar system :

Describing the solar system and its planetary arrangement showing position of the Earth in our solar system.

Our **Solar System** is a collection of planets, moons, asteroids, comets and smaller debris such as meteoroids which are revolving around the Sun. The Sun is the center of our Solar System. Everything in the Solar System orbits around the Sun. The order of planets, nearest to the farthest from the Sun, in our Solar System is:

- 1) Mercury
- 2) Venus
- 3) Earth
- 4) Mars
- 5) Jupiter
- 6) Saturn
- 7) Uranus
- 8) Neptune

Out of these eight planets, five are visible to the naked eye (Mercury, Venus, Mars, Jupiter and Saturn). The remaining two (Uranus and Neptune) require telescopes for their observation. Pluto is no more considered as a planet now; instead, it belongs to a new category called 'dwarf planet'.



10.2: The Solar System

Activity 1: Remembering the order of planets in our solar system.

A mnemonic is a sentence or phrase that helps us remember a set of words in a specific order. It uses the same first letter as the words we are trying to remember.

For example, a mnemonic for remembering the order of the planets is as follows:

Mercury	M	My
Venus	V	Very
Earth	E	Excellent
Mars	M	Mother
Jupiter	J	Just
Saturn	S	Served
Uranus	U	Us
Neptune	N	Nans

Try to make your own mnemonic.

The Sun

- 📘 Explain that Sun is a star.
- 📘 Differentiate between a star and planet.

We can see thousands of stars during a night, but not the Sun. On the other hand, the Sun is the only star that we see during daytime. The Sun not only provides us light and heat, but it also gives out dangerous ultraviolet light which may cause sunburn and skin cancer. However, without the Sun, there would be no daylight, and our planet would simply be a dark, frozen world, with no oceans of liquid water, and no life.

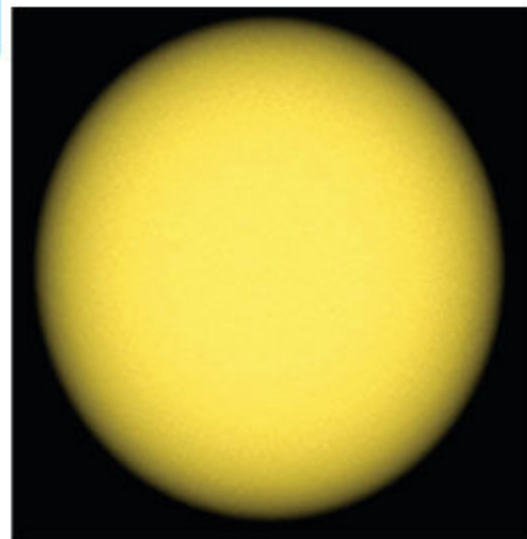


Figure. 10.3: The Sun

The Sun is a huge ball of hot gas, about 1.4 million kilometers across, which equals to 109 Earths if set side by side in terms of size. Also, it weighs as much as 330,000 Earths, and about 1,300,000 Earths would fit inside the Sun. The Sun lies about 150 million km away from us. At this distance, it takes about 8 minutes for the sunlight to reach us.

It is an extremely large star, however, in comparison with other stars; it is an average size yellowish star. Scientists estimate that the Sun was formed about 4.57 billion years ago. Our Sun is one of the billion stars orbiting the center of our galaxy.

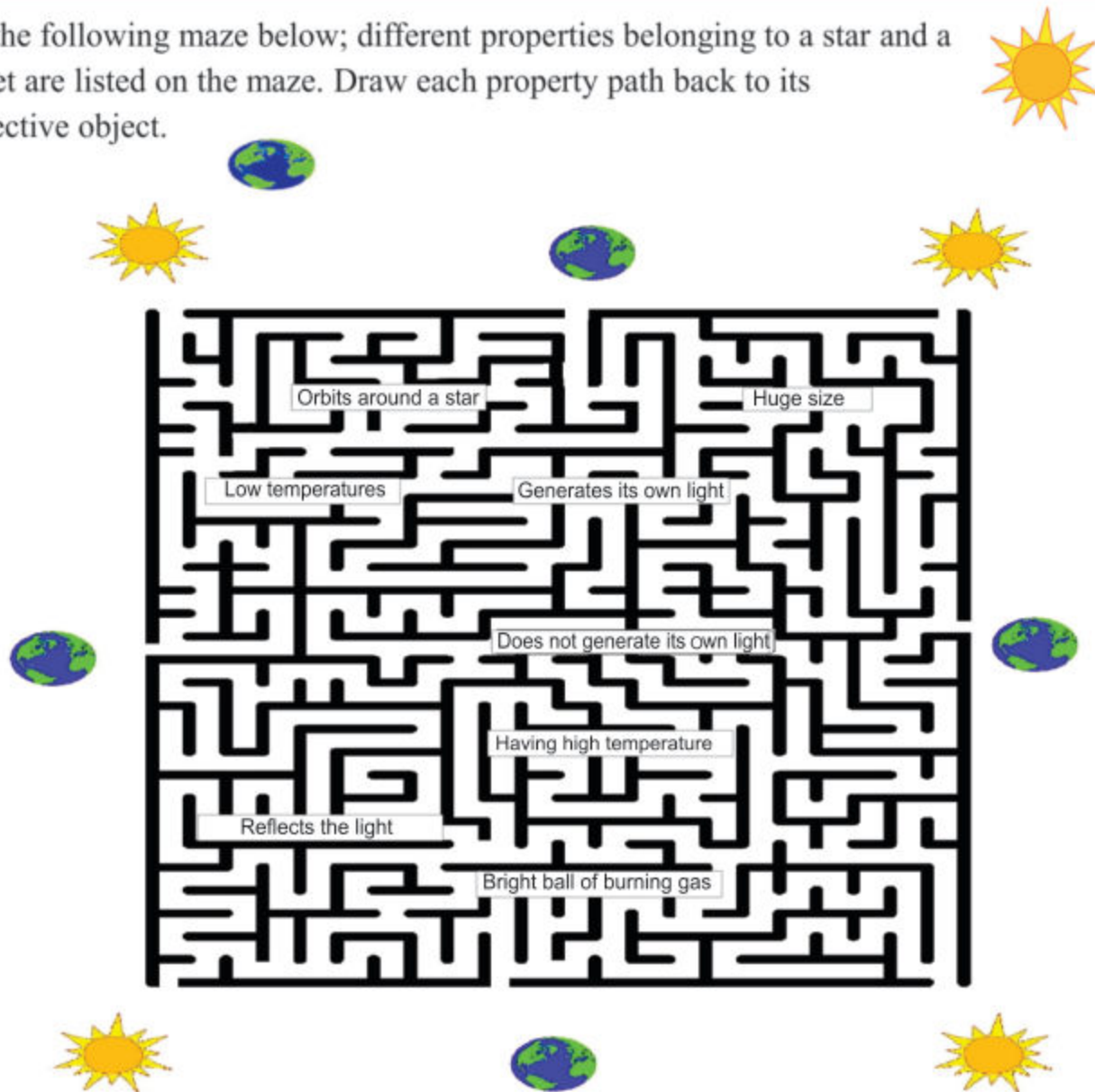
Let us understand the difference between a star and a planet.

A star, like the Sun, is a huge bright ball of burning gas that is held together by its own force of gravity. A star generates its own light. A star has very high temperature.

A planet, like Earth, is simply a large body that orbits around a star. A planet does not generate its own light, rather it reflects the light from the star it orbits. Planets have low temperatures.

Activity 2: Differentiating a star and a planet

See the following maze below; different properties belonging to a star and a planet are listed on the maze. Draw each property path back to its respective object.



The Planets:

- 1 Categorize planets and explain the relative size of the planets and their distance from the Sun.
- 1 Investigate the moons of the different planets of the solar system.

The planets in our Solar System are classified into two categories: **terrestrials** and **gas giants**

- **Terrestrial Planets: Mercury, Venus, the Earth and Mars** are all terrestrial planets. They are small and mainly made up of **rocks**.
- **Gas Giants: Jupiter, Saturn, Uranus and Neptune** are called **gas giants** because they are large and are made up of **gases**.

Size of the planets:

See figure 10.4(a), it shows the relative size (in the units of Earth radius) of each planet in comparison with the Earth. Each bar in the figure represents the size of the planet. Since Jupiter is the largest planet of the solar system, it has the largest bar in the figure. Similarly, Mercury has the smallest bar because it is the smallest planet in our solar system. The size of other planets can be observed likewise.

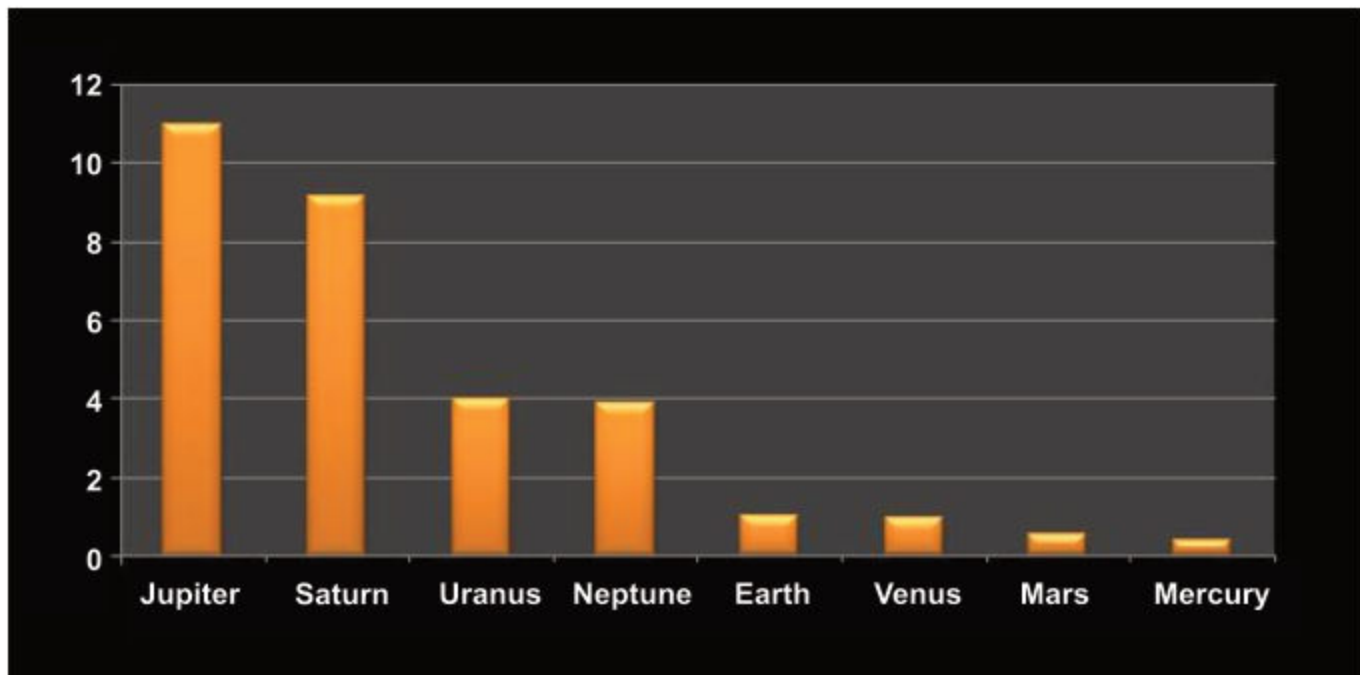


Figure 10.4: (a) Size of the planets in comparison with the Earth

Distance of the planets from the Sun:

Fig. 10.4 (b) represents the distance, in million (1000000) kilometers of the planets, as measured from the Sun. We observe that Mercury, being the first planet from the Sun, is the closest of all. As we move away from the Sun, the distance of planets increases rapidly.

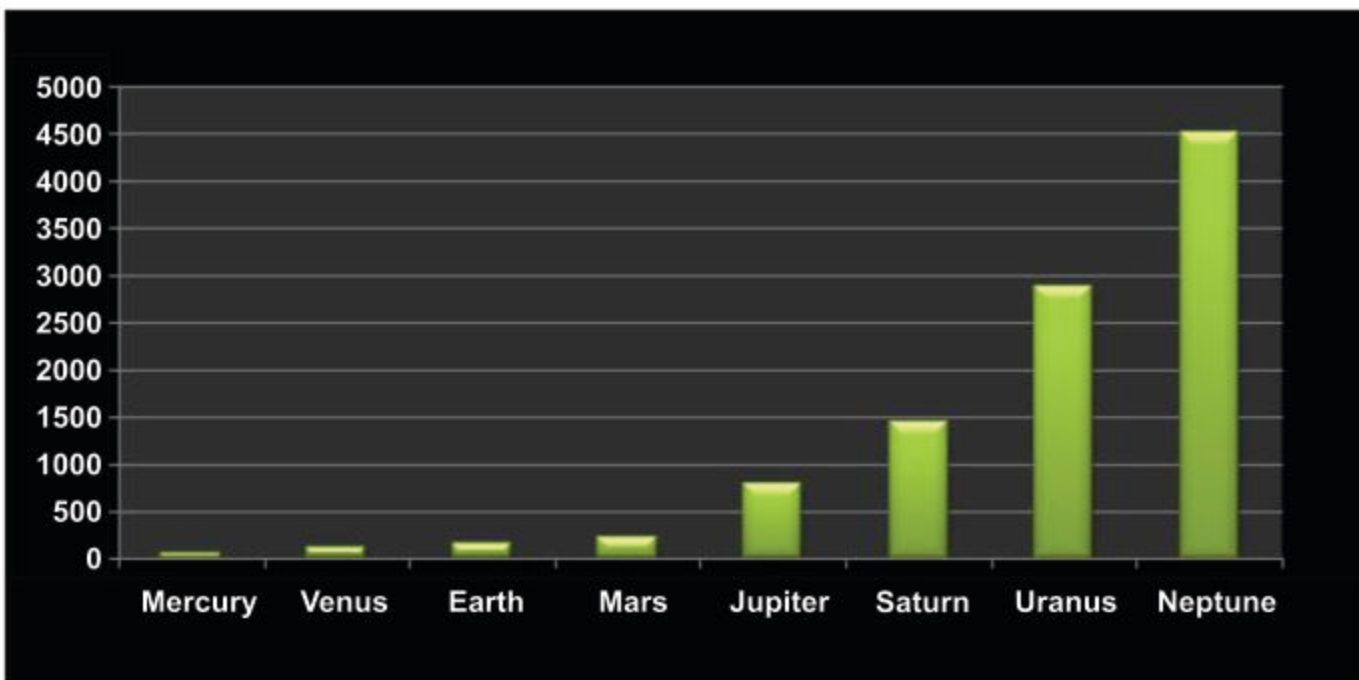


Figure 10.4: (b) Distance of the planets from the Sun

Let's have a closer look at each planet,

1. Mercury

Mercury is mostly made up of iron. The surface of the planet Mercury is covered with craters. Mercury has no atmosphere and it has no moons.

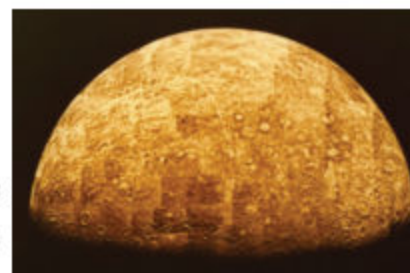


Figure 10.5: Mercury

2. Venus

Venus is the second brightest object in the night sky after the Moon. Like the Earth, Venus has an atmosphere. It is almost identical in size, chemistry and gravity, as the Earth. Venus has volcanoes, mountains and sand, just like Earth. It is sometimes called as "sister" of the planet Earth. But Venus has no moons.



Figure 10.6: Venus

3. Earth

The Earth is the biggest of all the terrestrial planets. Our planet is an oasis of life in an otherwise desolate universe. The Earth's temperature, weather, atmosphere and many other factors are just right to keep us alive. About 30% of the Earth's surface is covered with land, while about 70% is covered by oceans. The Earth has one moon and its name is Luna.



Figure 10.7: Earth

4. Mars

You may sometimes hear Mars referred to as the "Red Planet." This is because the surface of Mars is red. If you stood on the surface of Mars, you would see red dirt and rocks everywhere. Evidence suggests that Mars once had rivers, streams, lakes, and even an ocean.



Figure 10.8: Mars



Figure 10.9: Surface of Mars

As technology advanced, scientists were able to actually land a spacecraft on the surface of Mars. It has the largest volcano in the Solar System, which is known as "Olympus Mons". It has two moons; Deimos and Phobos.

5. Jupiter

Jupiter is a very stormy planet. The many different cloud formations and storms in the atmosphere make Jupiter a very colorful planet. Jupiter's great red spot is where a giant storm has been raging for at least 300 years. The spot at the lower right is also called "The Eye of Jupiter" because of its shape. Jupiter is considered a gas giant because it does not have a solid surface. Jupiter has 50 confirmed moons, while 12 moons are in the process of confirmation. One of its moons



Figure 10.10: Jupiter

“Ganymede” is the largest moon of our solar system. Ganymede, along with other three largest moons of Jupiter, were discovered by Galileo in 16th century.

6. Saturn:

In many ways, Saturn is similar to Jupiter and the second largest planet in our Solar System. Saturn is the least dense planet in our solar system. Like Jupiter, Saturn is wider in the middle and narrower near its top and bottom. Saturn is most well-known for its rings. The rings are split into categories, ranging from Ring A to Ring G. The rings are not solid but are rather made up of particles of ice, dust and rocks.

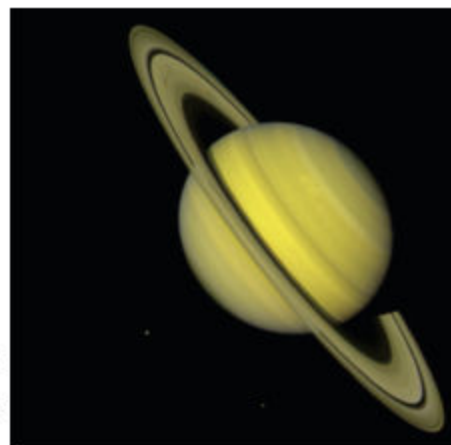


Figure 10.11: Saturn

Saturn has 53 confirmed moons and 9 still need confirmation. The most well-known of Saturn's moons is Titan. It is the second largest moon in the solar system next to Jupiter's Ganymede. Titan is larger than the planet Mercury.

7. Uranus:

Like Jupiter and Saturn, Uranus is a gas giant. But Uranus is a little different. Unlike all the other planets and most of the moons in our solar system, Uranus spins on its side. Uranus is an extremely cold planet. It has been called the "ice giant". Uranus is almost identical to the planet Neptune. Uranus has 27 moons. Five of these moons are large and the rest are smaller. The largest moon is Titania.

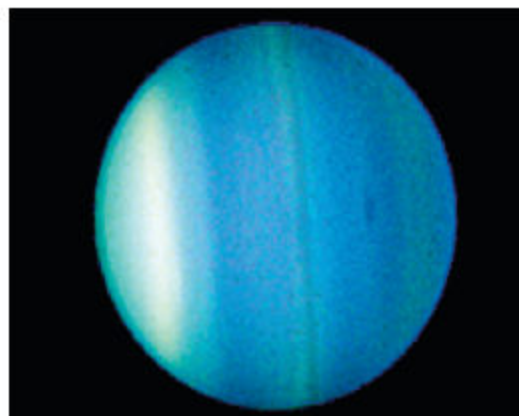


Figure 10.12: Uranus

8. Neptune:

Neptune is the smallest of the four gas giants in our Solar System. Not much was known about Neptune until it was visited by the spacecraft Voyager 2. Neptune is a very windy place. No other planet in the solar system has winds that are as strong as Neptune's. Neptune has six rings which circle the planet. Neptune has 13 moons that we know of. Because Neptune is so far away, it is difficult to see any of these.

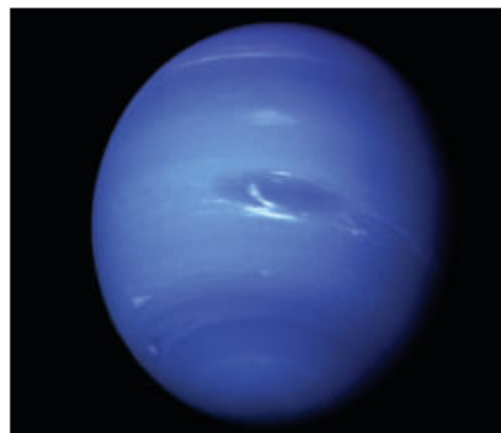
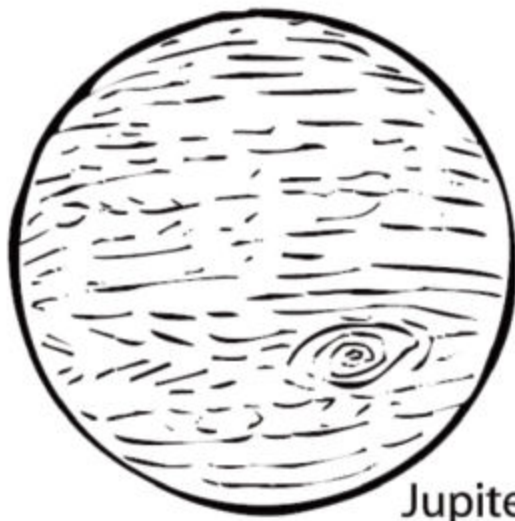


Figure 10.13: Neptune

Activity 3: Students should draw planets in their natural order in their copies. Also, colour the planets, as observed in the previous section.



Jupiter

Neptune



Mars

Earth

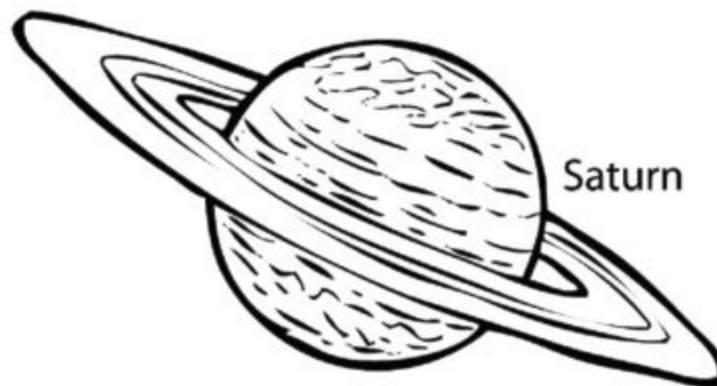


Uranus

○ Mercury



Venus



Saturn

Teacher Note: Teacher should guide the students to draw these pictures on drawing paper as home work.

Meteoroids, Asteroids and Comets

 Compare Meteoroids, Asteroids and Comets.

Meteoroids are rocky, stony and metallic debris in space revolving in a variety of orbits around the Sun at variety of speeds. They can be the size from a pebble up to several meters. When small meteoroids come close to the Earth, they get attracted towards it. They travel very fast and burn up because of the heat generated. They then produce a bright streak of light in the sky as they enter the Earth's atmosphere. We call them 'shooting stars' or more specifically 'meteors'. Large meteoroids can produce craters on the Earth's surface.



Figure 10.14 (a): Meteor

Asteroids are also rocky objects, mostly orbiting the Sun in a region between Mars and Jupiter which is known as 'the asteroid belt'. They can also produce craters on the Earth's surface. They come in size from a few meters up to several hundred meters. They are sometimes called 'Planetoids'. Vesta is an example of an asteroid.



Figure 10.14 (b): Vesta, an asteroid

Comets are small bodies made up of ice, dust and rock. They have been referred to as dirty snowballs. They have a center called 'nucleus' which is surrounded by a hazy cloud called 'coma'. They move in highly elongated orbits around the Sun. When they come close to the Sun, their gas evaporates due to the heat from the Sun, thus producing a long tail.

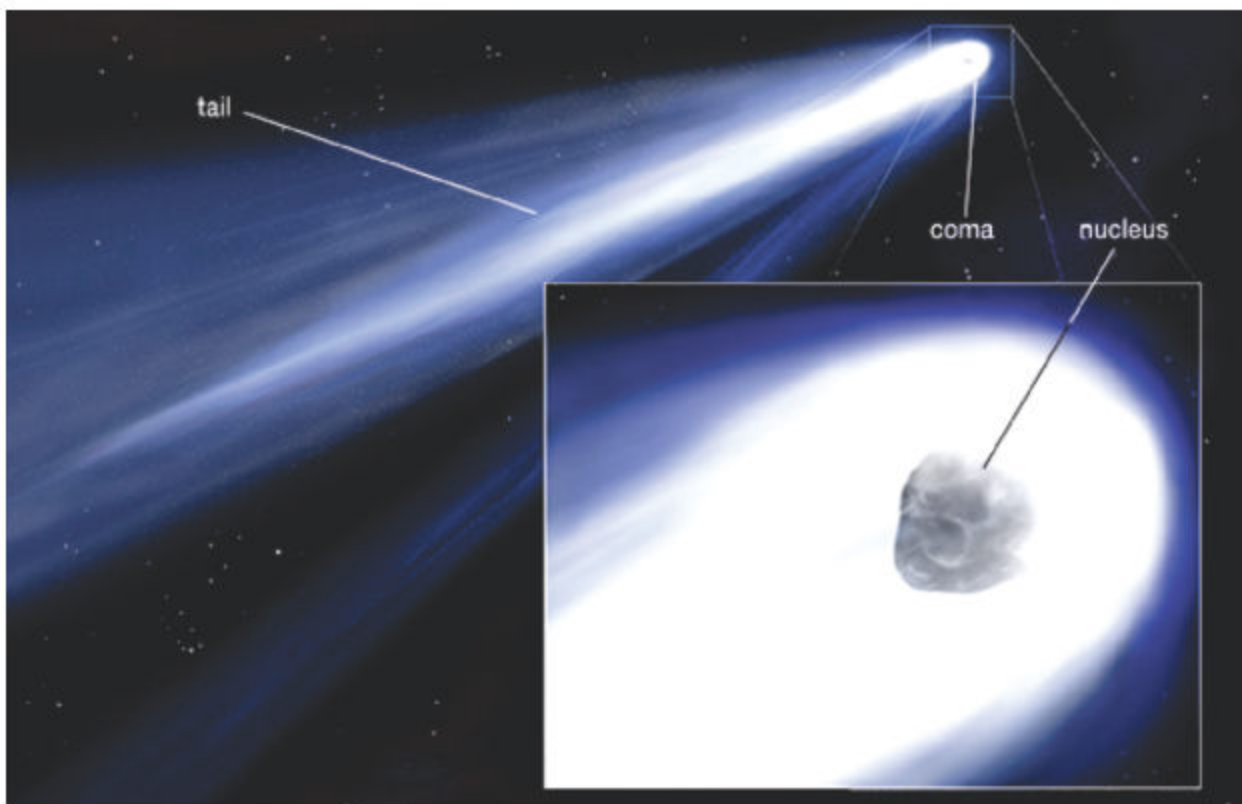


Figure 10.14 (c): Comet

Activity 4: Understanding the differences and similarities among meteoroids, asteroids and comets.

Pick the properties listed in the box below and write in front of the object they belong to. A property may belong to more than one object.

UP TO SEVERAL HUNDRED METERS LONG TAIL ICE, DUST AND ROCK
BURN UP IN ATMOSPHERE CAN PRODUCE CRATERS DIRTY SNOWBALLS
HAVE ELONGATED ORBITS COMA SHOOTING STARS PLANETOIDS
PRODUCE A BRIGHT STREAK OF LIGHT VARIETY OF ORBITS NUCLEUS
UP TO SEVERAL METERS ROCKY, STONY AND METALLIC
FOUND REGION BETWEEN MARS AND JUPITER

Comets:



Asteroids:



Meteoroids:



Review questions:

1) Read and complete the sentences with appropriate words given below:

- i. ----- is a collection of planets, moons, asteroids, comets and smaller debris such as meteoroids that are revolving around the Sun.
- ii. The arrangement of first four planets is -----, -----, ----- and -----.
- iii. Out of these eight planets, ----- are visible to the naked eye.
- iv. ----- is now considered as dwarf planet.
- v. ----- is the brightest and the largest object in our solar system .
- vi. The Sun is a glowing ball of -----.
- vii. -----does not generate its own light.
- viii. Planets in our solar system can be classified into two categories: ----- and -----.
- ix. ----- is the largest planet of our solar system .
- x. ----- is the farthest planet in our solar system .
- xi. Mars is also referred to as -----
- xii. The most remarkable feature of Jupiter is its great -----.
- xiii. ----- is a fragment of rock that survives the journey through the Earth's atmosphere
- xiv. ----- is a rocky object smaller than a planet orbiting the Sun.
- xv. ----- are referred to as dirty snowballs.

Neptune	Mercury	gas giants	Venus	the Earth	red spot
Saturn	The Sun	red planet	Solar system	five	Pluto
meteorite	asteroid	Mars	Jupiter	comets	terrestrial
					gas Planet

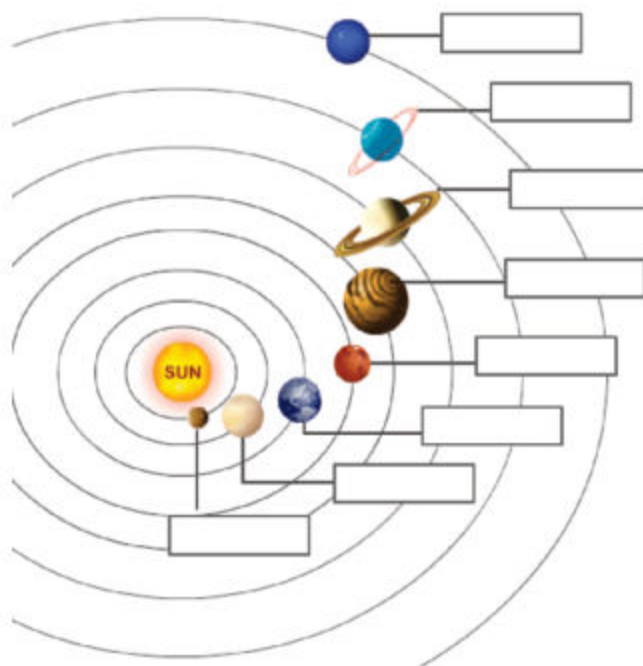
2) Circle the **True** and **False** statements and write down the correct statement for False statements:

- | | | | |
|-------|---|----------|----------|
| i. | Mercury is the brightest planet in the sky. | T | F |
| ii. | Venus has two moons. | T | F |
| iii. | Pluto is the ninth planet of our solar system . | T | F |
| iv. | Everything in the solar system is orbiting the Sun. | T | F |
| v. | The Sun provides the light and heat necessary to sustain life on the Earth. | T | F |
| vi. | Mars is called the sister of the planet Earth | T | F |
| vii. | Jupiter is the second biggest planet of the solar system . | T | F |
| viii. | Asteroid belts lie between Jupiter and Saturn . | T | F |
| ix. | Comets have bright tails which point away from the Sun. | T | F |
| x. | Saturn has a system of rings, ranging from A to G. | T | F |

3) Choose the correct answer from the given options:

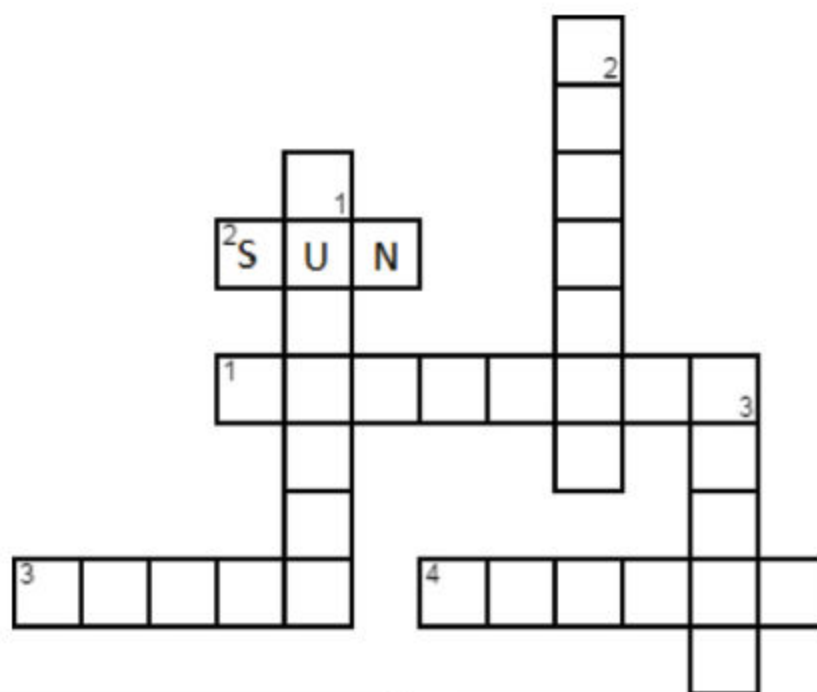
- i. ----- generate their own light.
a. planets b. stars c. dwarf planets d. asteroids
- ii. Our sun is one of the -----of stars in our galaxy.
a. millions b. billions c. hundreds d. thousands
- iii. Terrestrial planets are mainly made up of -----
a. ice b. water c. rocks d. carbon material
- iv. Mars has ----- moon(s).
a. one b. three c. two d. none
- v. The planet mostly made up of iron is -----
a. venus b. mars c. mercury d. earth
- vi. Venus and Earth are identical in -----
a. size b. chemistry c. gravity d. all of above
- vii. The largest volcano of solar system is “Olympus Mons” lies in -----
a. mars b. jupiter c. earth d. uranus
- viii. Neptune has -----
a. five rings b. six rings c. four rings d. eight rings
- ix. The center of a comet is called -----
a. coma b. nucleus c. tail d. ice
- x. It takes ----- minute for light to reach the earth
a. 6 b. 10 c. 8 d. 12

4) Identify and fill in the names of the planets:



- 5) Complete the puzzle with the help of the hints given below. Use capital letters.
Answers are given at the end of the chapter.

The Solar System



Across

1. The largest moon of the solar system
2. Center of our solar system
3. Now called the Dwarf Planet
4. Known as Dirty snowballs

Down

1. Name of Pakistan's national space agency
2. The largest planet
3. The only Planet with life

Key

Across

1. Ganymede
2. Sun
3. Pluto
4. Comets

Down

1. SUPARCO
2. Jupiter
3. Earth